Abel Magnago

Software Engineer | AI systems in production (RAG, guardrails, evals) | Node.js · TypeScript · AWS

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Summary

Software engineer with 5+ years shipping systems end to end, from public-sector audit and legal tech to an e-learning platform used by 30,000+ students. I built the retrieval and guardrail layer of a production AI assistant that answers questions over organizational documents: chunking, vector search in Qdrant, citations back to the source file, context management and input hardening. Backend in Node.js, TypeScript and Python, React on the frontend, AWS for deploy. Remote from Brazil (UTC-3), with overlap for EU and US East.

Technical Skills

Languages: TypeScript, JavaScript, Node.js, Python, PHP, SQL, HTML5, CSS3

AI & Retrieval: RAG, embeddings (OpenAI), Qdrant vector search, chunking and reindexing, context and token budgeting, prompt/tool-call validation (Ajv), jailbreak filtering, rate limiting, MCP, agent tooling

Frontend: React, Next.js, TypeScript, Tailwind CSS, Radix UI, Styled-components, Storybook, EJS, Vite

Backend & APIs: Node.js, Express, REST APIs, OpenAPI/Swagger, WebSockets (Socket.IO), SOAP, JWT, OAuth, LDAP/SSO, Zod and Joi validation, Firebase

Databases: PostgreSQL, MySQL, MongoDB (Mongoose), DynamoDB, data modeling

Cloud & DevOps: AWS (Lambda, API Gateway, S3, CloudFront, Route 53, ACM, Cognito, DynamoDB, SQS, CloudFormation), AWS SDK v3, Serverless Framework, Docker, Docker Compose, GitHub Actions, CI/CD, Git, GitHub, Bitbucket

Testing & Quality: Jest, Supertest, unit and integration testing against real databases (mongodb-memory-server), ESLint, Prettier, Husky, lint-staged, commitlint, pnpm, esbuild

Practices: code review, pair programming, agile/Scrum, backlog management, clean architecture, dependency injection, design systems, web security (Helmet, rate limiting, bcrypt), data protection (LGPD/GDPR)

Experience

Software Engineer — *Rio de Janeiro, Brazil*
Controladoria-Geral do Estado do Rio de Janeiro (CGE RJ) — *May 2023 – present*
Started on a contract basis in May 2023 and moved to full-time in March 2025.

- Design, build and maintain the internal systems state auditors and administrative staff use every day, in workflows where a wrong number in a report is a real problem.
- Built a document signer with SHA-256 hashing so a signed file can be verified as untampered, which is what public institutions need to treat it as legally valid.
- Internal events platform where any civil servant can register, attend and download a training certificate, plus a logistics system that tracks assets and requests between departments.
- Added LDAP/SSO login with sessions persisted in MongoDB, replacing separate credentials per system.
- Led automation and data integration work between departments, removing manual handoff steps that used to consume hours of staff review.
- Built the retrieval and guardrail layer of an internal AI assistant that answers questions over organizational documents: sliding-window chunking with overlap, Qdrant vector search, citations back to the source file, token-budgeted context trimming, and request validation that rejects disallowed models, oversized payloads and known jailbreak patterns in English and Portuguese.
- Migrated the legacy PHP audit system to Node.js, Express and TypeScript with React, PostgreSQL, MySQL and MongoDB. It is still in use today, and I later optimized it for performance and maintainability.
- Own projects end to end, from requirements and data modeling through the API, frontend and deploy, and stay on production incidents until the fix is live.

Software Consultant (Contract, concurrent) — *Rio de Janeiro, Brazil*
Extreme Digital Solutions (EDS) — *Mar 2025 – Sep 2026*

- Delivered full-stack features for legal and audit clients in PHP, Python, Node.js, TypeScript and React, taking them from requirements to production.
- Was the technical point of contact on several engagements: gathered requirements, translated business rules into working software, and explained the trade-offs in plain language.
- Integrated legacy platforms with modern web applications so clients kept their existing data instead of paying for a risky full rewrite, and refactored legacy codebases and slow database queries.
- Automated processes that depended on manual spreadsheets and file handling, and set up Docker environments and CI/CD pipelines so builds and deploys stopped being manual events.

Front-End Developer (Contract, concurrent) *Brazil*
ProRaiz Sistema de Ensino *Jun 2024 – Dec 2025*

- Led front-end work on a learning platform used by more than 30,000 active students, managing the backlog and the delivery pipeline from sprint to release.
- Migrated legacy modules onto React, TypeScript and Next.js, cutting build times by 40% and improving the experience for those 30,000+ users on slow connections and low-end devices.
- Worked with design and product to turn UX requirements into accessible, performant React components.
- Mentored interns and junior developers through pair programming and code review, and documented team conventions so the same mistakes stopped repeating.

Front-End Developer *Rio de Janeiro, Brazil*
Prodígio Educação *Jun 2022 – Mar 2025*

- Built applications for students, teachers and content managers across several verticalized products, each with its own rules and data model.
- Shared modules and components across products with React, TypeScript and Firebase, reducing duplicated code between teams.
- Moved legacy front-end modules onto the modern React stack one module at a time, without breaking anything in production.

QA Intern / Front-End Contributor *Rio de Janeiro, Brazil*
Prodígio Educação *Oct 2021 – May 2022*

- Owned high-traffic landing pages for e-learning products, building new ones as campaigns launched and keeping load speed and cross-browser rendering consistent.
- Ran functional testing on each release and documented defects before they reached students, and used Storybook to validate component states.

Certifications

AWS Knowledge Training Badges (7): Serverless, Compute, Networking Core, Object Storage, Data Migration, Cloud Essentials, AI Driven Development Lifecycle Foundations (Amazon Web Services, Oct 2026)

AI and Digital Transformation in Government: Saïd Business School, University of Oxford (Oct 2026)

AWS Essentials: JStack (Mar 2025)

Education

Universidade Unigranrio | Afya *Rio de Janeiro, Brazil*
Associate's Degree (CST) in Systems Analysis and Development *Aug 2021 – Dec 2023*

Languages

Portuguese (native) · English (B1, actively improving)